Listen First, Then Answer: Timestamp-Grounded Speech Reasoning

Anonymous submission to Interspeech 2026

Abstract

Large audio-language models (LALMs) can generate reasoning chains for their predictions, but it remains unclear whether these reasoning chains remain grounded in the input audio. In this paper, we propose an RL-based strategy that grounds the reasoning outputs of LALMs with explicit timestamp annotations referring to relevant segments of the audio signal. Our analysis shows that timestamp grounding leads the model to attend more strongly to audio tokens during reasoning generation. Experiments on four speech-based benchmark datasets demonstrate that our approach improves performance compared to both zero-shot reasoning and fine-tuning without timestamp grounding. Additionally, grounding amplifies desirable reasoning behaviors, such as region exploration, audiology verification, and consistency, underscoring the importance of grounding mechanisms for faithful multimodal reasoning. [1]

Index Terms: Large Audio Language Models, Reasoning, Grounding, Interpretability

1. Introduction

Large audio-language models (LALMs) [1–5] have recently shown promising performance across a wide range of audio understanding and reasoning tasks. However, despite their multimodal design, most existing LALMs lack *explicit mechanisms for grounding* their reasoning in concrete acoustic evidence. As a result, model predictions are often driven by abstract linguistic priors rather than faithful engagement with the input audio signal. This disconnect between reasoning and acoustic evidence can lead to fragile Chain-of-Thought (CoT) [6–8] and compromise the faithfulness and interpretability of the explanations.

In this paper, we hypothesize that LALMs both "hear better" and "reason better" when their textual reasoning steps are explicitly grounded in precise temporal segments of the audio signal. This hypothesis is inspired by human auditory cognition, where listeners dynamically shift their attentional focus over time to selectively attend to task-relevant acoustic events during reasoning and decision-making [9, 10]. As shown in Fig. 1, timestamp-based grounding promotes more targeted and systematic cross-referencing between linguistic reasoning and acoustic evidence throughout multi-step inference. Hence, we utilize text-to-timestamp object-centric grounding as the intermediate reasoning stage, where the predicted timestamp zones with explanation serve as simple but effective CoT signals to help improve the quality of the final reasoning step.

Our contributions can be summarized as follows:

[1] Code and audio examples are available at an anonymous repository: https://anonymous.4open.science/r/Timestamp-Grounded-Speech-Reasoning-5341

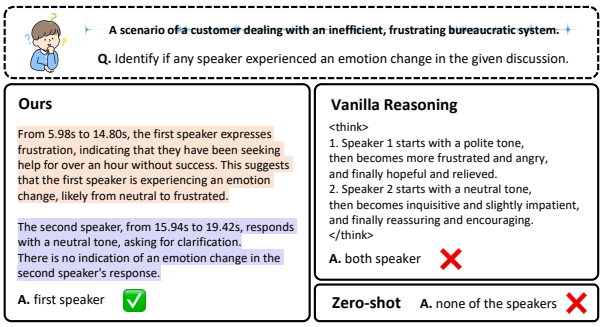

Figure 1: ***Grounded timestamp reasoning enables interpretable and accurate answers.*** *Our method decomposes the task into a sequence of reasoning steps explicitly grounded in corresponding audio regions. In contrast, vanilla reasoning and zero-shot tend to produce ungrounded and incorrect responses.*

- We identify and analyze a fundamental limitation of current LALMs, showing that correct predictions do not necessarily imply faithful audio grounding, and that existing models often rely on text-biased reasoning with weak engagement of acoustic evidence.
- We propose a timestamp-grounded reasoning framework that anchors reasoning steps to temporally localized audio segments, with a two-stage strategy that combines supervised timestamp alignment and reward-based optimization.
- We present comprehensive evaluations across multiple speech benchmarks and behavior-level analyses, demonstrating that timestamp grounding improves not only end-task accuracy but also reasoning consistency, audio verification behavior, and overall interpretability.

2. Related Works

Reasoning in Audio-Language Models. CoT reasoning has emerged as an effective paradigm for improving the transparency and accuracy of large language models by modeling intermediate reasoning steps [11–14]. This paradigm has been extended to LALMs, where embedded reasoning integrates linguistic abstractions, including models such as the Thinker variants of the Qwen family [15, 16] and Audio Flamingo 3 [17]. Recent studies aim to enhance reasoning quality, showing that it can be achieved even with lightweight models [18], while works such as Audio-CoT [19] and Audio-Reasoner [20] further investigate explicit and structured CoT pipelines via standard fine-tuning. Moreover, reinforcement learning has been adopted to further enhance reasoning fidelity, including R1-AQA [21], Omni-R1 [22], and Audio-Thinker [23], which fine-tune open-

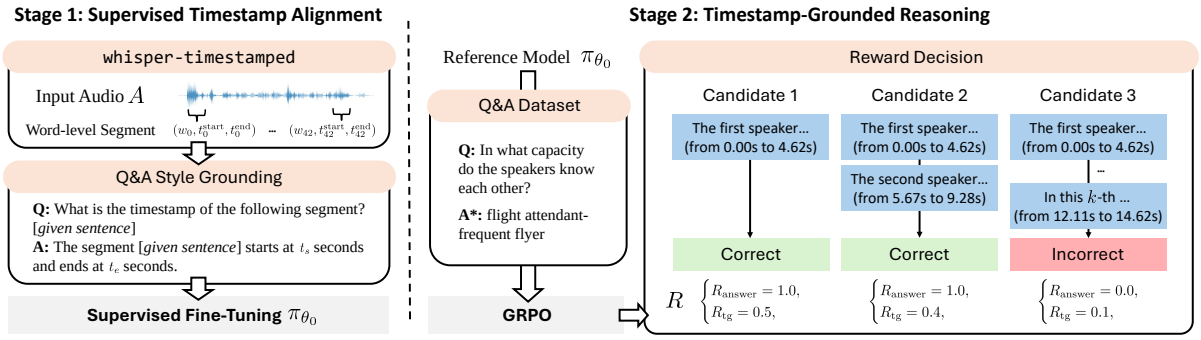

Figure 2: ***Overview of the proposed framework.*** *Stage 1 learns temporal localization through supervised timestamp alignment. Stage 2 performs GRPO-based optimization to encourage answers justified by concise timestamp-grounded reasoning.*

weight audio models using Group Relative Policy Optimization [24]. While these approaches achieve notable gains in reasoning accuracy, they primarily *emphasize performance improvements rather than explicitly assessing whether reasoning steps are grounded in actual audio evidence.*

Grounded Reasoning for Multimodal Language Models. Recent studies have explored grounded reasoning as a means to promote cross-referencing between textual and non-textual modalities. In the vision-language domain, prior works such as Visual-RFT [25], MM-GCoT [26], ARGUS [27], and ViGoRL [28] integrate visual signals, most commonly object bounding boxes, into the reasoning process, thereby improving faithfulness between reasoning steps and perceptual evidence. In contrast, grounded reasoning in LALMs remains largely underexplored. Step-Audio-R1 [29] represents an initial attempt to introduce audio grounding into reasoning; however, its grounding is not quantitatively measured, and the presence of acoustic references is determined via heuristic criteria. Relatedly, Timestamped Audio Captioner (TAC) [30] proposes temporally grounded audio captioning, generating time-aligned descriptions. However, TAC itself is not a reasoning model; rather, it serves as a captioning system whose outputs are provided as inputs to downstream LLMs. By contrast, our work directly addresses these limitations *by internally assessing whether real audio evidence is utilized, and by strengthening the explainability of CoT through auditory interactions.*

3. Methodology

Our goal is to equip the reasoning chains generated by LALMs with timestamps that correspond to salient segments of the input audio. To this end, we propose a two-stage training framework that progressively builds robust timestamp grounding capabilities within the reasoning. In the first stage, the model is trained solely on timestamp prediction; in the second stage, timestamp prediction is integrated into downstream QA reasoning tasks. The overall structure is represented in Fig. 2.

3.1. Stage 1: Supervised Timestamp Alignment

A key obstacle to faithful audio reasoning is that most existing LALMs are not reliably capable of extracting temporally precise evidence. In practice, timestamp references are often missing, coarse, or misaligned with the true acoustic speech events, which can propagate to erroneous CoT. Therefore, before optimizing high-level reasoning behaviors, we first establish a strong temporal grounding primitive through supervised timestamp alignment (STA).

Timestamped supervision. To construct timestamp supervision at scale, we build a timestamp-annotated speech corpus using `whisper-timestamped` [31], an extension of Whisper [32] that produces word-level timestamps and confidence scores. Unlike vanilla Whisper, which primarily provides segment-level timestamps, `whisper-timestamped` estimates word timestamps by applying Dynamic Time Warping [33] to cross-attention weights, enabling finer and more reliable temporal localization.

From timestamped transcript to Q&A-style grounding data. Given an audio file A, we first obtain a word-level timestamped transcript $\{(w_j, t_j^{\text{start}}, t_j^{\text{end}})\}_j$. We then convert the timestamped transcript into question-answer instances designed to train explicit temporal anchoring, as shown in Fig. 2. Specifically, we sample a text phrase from the transcript as a query and ask the model to predict the corresponding temporal region in input audio A, where [*given sentence*] indicates a spoken sentence whose start (t_s) and end (t_e) times are to be grounded. This method trains the model to generate start-end timestamps, making it compatible with our timestamp-grounded reasoning framework and producing a reference model, denoted as π_{θ_0}.

3.2. Stage 2: Timestamp-Grounded Reasoning

With an established textual reasoning foundation, we now address the core challenge of equipping the model's reasoning process with explicit timestamp grounding. We adopt *Group Relative Policy Optimization (GRPO)* [24], a reinforcement learning framework that stabilizes policy learning by leveraging relative comparisons among multiple sampled trajectories.

Answer Correctness Reward (R_{answer}). The first component, R_{answer}, measures whether the model produces the correct multiple-choice answer. We extract the predicted option from the final line and assign a binary reward:

$$R_{\text{answer}} = \begin{cases} 1.0, & \text{if pred. matches GT.} \\ 0.0, & \text{otherwise.} \end{cases} \tag{1}$$

This term ensures that reward-based refinement preserves task-level accuracy and encourages the generated reasoning to remain grounded in the final answer.

Timestamp Grounded Reward (R_{tg}). While R_{answer} evaluates *what* answer the model produces, it does not assess *how* the answer is justified. To promote concise and meaningful timestamp-grounded reasoning, we introduce a timestamp grounding reward R_{tg}, which evaluates whether the generated reasoning refers to relevant audio timestamps and encourages compact timestamp usage. This design is aligned with prior

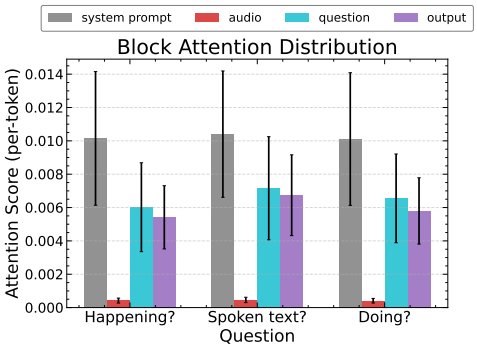

Figure 3: ***Semantic attention distribution of a baseline LALM across input categories.*** *It aggregates token-level attention by semantic role, highlighting under-attention to audio features.*

works [23,34,35] on effective reasoning that emphasizes reducing unnecessary steps while preserving semantic evidence.

We identify distinct grounding units that explicitly reference temporal segments of the input audio. Let k denote the *number of timestamp-grounded units* in the generated completion. The compaction score R_{tg} is defined as:

$$R_{\text{tg}} = \begin{cases} 0, k = 0; \quad C_{\max}, k \leq k_{\text{ref}}; \quad C_{\min}, k \geq k_{\max}; \\ C_{\max} - (k - k_{\text{ref}}) \frac{C_{\max} - C_{\min}}{k_{\max} - k_{\text{ref}}}, \quad \text{otherwise.} \end{cases} \tag{2}$$

where k_{ref} denotes the reference number of timestamped reasoning units (set to 1 in our experiments), and $C_{\max}$ and $C_{\min}$ denote the maximum and minimum compaction rewards (set to 0.5 and 0.1 respectively). This formulation encourages concise yet sufficiently grounded reasoning, avoiding both missing grounding signals ($k = 0$) and overly verbose timestamp usage.

Final Reward. The final reward is computed as:

$$R = R_{\text{answer}} + R_{\text{tg}} \tag{3}$$

This formulation encourages the model to not only predict the correct answer, but also justify it using concise, temporally grounded evidence, as shown in Fig. 1, 2.

4. Experimental Settings

4.1. Datasets

We use a mixture of timestamp-annotated speech datasets for Stage 1 training, including LibriSpeech [36], CoVoST 2 [37], MELD [38], multi-speaker [20], and YouTube8M [39] speech data. The resulting corpus contains approximately 268k examples, which we plan to release to facilitate future research. For Stage 2, we perform reward-based optimization using question-answering datasets, including MELD [38], multi-speaker dialogue QA [20], and YouTube8M speech QA [39], resulting in approximately 47k training examples.

4.2. Implementation Details

We conduct our experiments using Qwen2.5-Omni [40] and Audio Flamingo 3 [17] as the base models. Training is performed on a single node with 8 H200 GPUs. We use a per-GPU batch size of 8 with 4 gradient accumulation steps. The learning rate is set to 2×10^{-5} for Stage 1, 5×10^{-6} for Stage 2, with a sampling temperature of 0.8, 8 responses per GRPO step, and a KL regularization coefficient $\beta = 0.04$.

Table 1: *Effect of Sentence-level Timestamp Alignment.*

Model	STA	IoU ↑	F1 ↑	IoU ≥ 0.7 (%) ↑
Qwen-2.5-Omni [40]	✗	0.2324	0.3074	11.32
	✓	0.7189	0.7675	69.79
AudioFlamingo3 [17]	✗	0.2838	0.3592	11.45
	✓	0.5439	0.6012	47.95
Gemini-2.5-Flash [44]	–	0.7356	0.8141	71.08

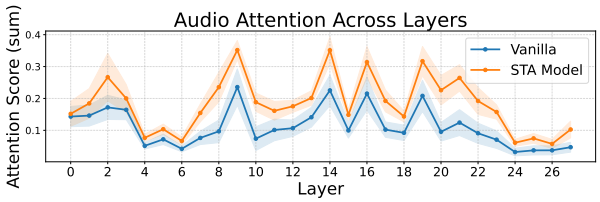

Figure 4: ***Layer-wise audio attention during timestamp generation.*** *STA model allocates higher attention to audio tokens across transformer layers compared to the vanilla model, indicating stronger reliance on acoustic evidence.*

5. Results

5.1. On the Limited Use of Audio Information in LALMs

Despite their multimodal design, we observe that current LALMs exhibit a pronounced text-biased reasoning tendency, closely resembling previously reported in vision-language models [41, 42]. To analyze this behavior more systematically, we conduct a **semantic-based attention analysis** that aggregates attention weights over system tokens, audio tokens, instruction tokens, and self-referential outputs. We perform this analysis on the MMAU [43] speech data using audio-critical queries, {*What is happening?*, *What is the spoken text?*, and *What are they doing?*}, for which the answer cannot be inferred from textual instructions alone and requires access to the audio signal.

As shown in Fig. 3, baseline LALMs allocate only a small fraction of attention to audio tokens, even when producing correct or seemingly audio-grounded answers. We observe a clear attention sink effect: system tokens dominate attention allocation even on a per-token basis, receiving over 15× more attention than audio tokens. These findings indicate that successful predictions do not necessarily imply faithful audio reference. As a result, models may generate plausible yet weakly grounded explanations that do not reflect the true causal role of audio in the reasoning process. This behavior motivates the need for training strategies that explicitly encourage alignment between reasoning steps and grounded audio evidence.

5.2. Grounding is Listening

In this section, we analyze *whether LALMs genuinely rely on audio evidence during grounding* and evaluate the effectiveness of timestamp supervision. Before we dive in, STA improves timestamp grounding accuracy, as reflected by consistent gains in Table 1, while the vanilla model shows timestamp hallucinations. We report the intersection-over-union (IoU) between predicted timestamp ranges and ground-truth segments, the proportion of predictions achieving high-overlap alignment (IoU $\geq$ 0.7), and Sound Event Detection F1 score (SED-F1) [45], which measures event-level temporal precision. Across all metrics, our method outperforms the vanilla baseline and achieves performance comparable to Gemini-2.5-flash [44], indicating robust temporal localization.

Table 2: *Performance comparison on MMAU-mini-Speech, MMAR-Speech, AIR-Bench, and MELD.*

Methods	Size	MMAU-mini	MMAR	AIR-Bench			MELD
		Speech (%)	Speech (%)	SER (%)	SNV (%)	SIC (%)	(%)
Proprietary Models							
Gemini 2.5 Flash [44]	–	**75.08**	**72.11**	56.4	68.5	88.6	61.5
GPT-4o Audio [48]	–	66.67	70.41	51.2	61.6	**89.3**	62.5
Open-source Models							
SALMONN [3]	7B	26.43	24.35	29.9	34.3	42.3	37.2
Audio Flamingo 3 [17]	7B	66.37	57.48	59.5	**76.8**	79.6	58.5
Audio Reasoning Methods							
Audio-CoT [19]	8.4B	55.26	34.01	–	–	–	–
Audio-Reasoner [20]	8.4B	66.07	32.99	60.5	56.3	88.1	63.2
Audio-Thinker [23]	8.4B	73.37	64.29	56.2	67.5	–	–
Ablation Variants							
Qwen2.5-omni [40]	7B	70.60	59.86	60.2	63.9	83.5	60.3
+ Only STA	7B	71.17	61.22	59.5	66.0	84.3	62.8
+ Reasoning SFT	7B	74.47	62.93	58.5	68.1	85.0	61.8
Ours	**7B**	74.47	64.63	**62.5**	70.4	**89.3**	**64.6**

Next, Fig. 4 provides empirical evidence supporting the claim that **grounding is listening**. We compare semantic audio attention maps between the vanilla and the timestamp-aligned model, focusing specifically on the *time generation* phase, where the model is required to produce explicit temporal references. The timestamp-aligned model allocates higher attention to audio tokens across layers during time generation, suggesting that improved grounding is closely tied to faithful listening to temporally relevant audio cues.

Notably, we observe that certain layers exhibit a pronounced shift in listening behavior, acting as effective *picking points* where acoustic evidence is selectively amplified. This observation is consistent with prior analyses of the LLM domain [46, 47], which indicate that different layers internalize distinct functional roles. From this perspective, STA appears to reshape the behavior of layers that integrate external evidence, leading to more faithful incorporation of audio cues into the model's internal representations.

5.3. Timestamp Grounding Improves Benchmark Ability

Table 2 compares our method with ***Proprietary*** and ***Open-source*** LALMs on four representative benchmarks: MMAU-mini-Speech [43], MMAR-Speech [49], AIR-Bench [50] and MELD [38]. These benchmarks evaluate speech understanding, speech reasoning, and dialog-level recognition, providing a comprehensive assessment of speech-centric ability.

Starting from the Qwen2.5-Omni [40] baseline, we first analyze the impact of individual components through an ***Ablation Variants***. Applying only STA shows minor improvements, suggesting that it does not fully resolve domain alignment. We further evaluate a reasoning SFT variant *without timestamp supervision*, which improves over the baseline on several benchmarks but underperforms our model. These results indicate that timestamp alignment and reasoning supervision alone are each insufficient, highlighting the need to integrate temporal grounding with reasoning to achieve robust performance.

Our full model achieves the best overall performance across all benchmarks. In particular, it obtains the strongest results on AIR-Bench and MELD, even if compare with ***Proprietary Models***, demonstrating robust speech understanding and integration of acoustic cues. Moreover, on speech reasoning benchmarks such as MMAU-mini-Speech and MMAR-Speech, our method outperforms existing ***Audio Reasoning Methods***. Despite being designed for reasoning, prior audio reasoning methods show limited improvements on these benchmarks, whereas our model achieves stronger performance. This observation indicates that grounding-based supervision provides a more effective foundation for speech reasoning.

Table 3: *Average visual reasoning behaviors per example on MMAU-mini-Speech.*

Model	Regions Explored	Audiology Verify ↑	Consistency ↑	Acc. ↑
Zero-Shot				
Standard CoT	1.3	0.27	0.72	69.4
RL-tuned				
w/o compaction	3.6	0.48	0.78	72.8
w/o grounding	0.5	0.22	0.69	74.2
Ours	1.8	**0.56**	**0.83**	**74.5**

Table 4: *Average audio attention behaviors when generating reasoning on MMAU-mini-Speech.*

Model	Vanilla	Only STA	Reasoning SFT	Ours
Audio Attention (All)	0.0748	0.0936	0.0975	**0.1030**
Reasoning Attention	0.0790	0.0952	0.1032	**0.1138**
Answer Attention	0.0421	0.0690	**0.0714**	0.0680

5.4. Timestamp Grounding Improves Behavior Quality

Table 3 goes beyond end-task accuracy and examines how timestamp grounding reshapes the model's reasoning behavior. *Regions Explored* measures the number of distinct timestamped reasoning segments produced by the model. *Audiology Verify* evaluates the alignment between timestamped explanations and their corresponding audio segments using a Whisper-based similarity score [32]. Finally, *Consistency* measures the logical alignment between the model's reasoning trace and its final prediction, evaluated by a judge model (Qwen3-32B [16]).

Compared to the zero-shot baseline, ***RL-tuned variants*** show noticeable changes. The model w/o compaction explores more regions and increases audio verification, but achieves lower accuracy. In contrast, the variant w/o grounding achieves reasonable accuracy yet shows minimal region exploration and weak audio verification, suggesting that objective-driven training encourages shortcut strategies. Our full model achieves the highest audio verification and the strongest reasoning consistency, although it explores fewer regions.

Table 4 reports the average audio attention allocated to audio tokens across the ***Ablation Variants*** introduced in Table 2. Compared to the vanilla zero-shot reasoning baseline, all ablation variants allocate more attention to audio inputs during reasoning, suggesting that grounding signals and reasoning supervision encourage the model to rely more on acoustic evidence when forming reasoning chains. Despite this general increase, our full model achieves the highest audio attention overall, with the most pronounced gains observed during the reasoning stage. These results indicate that timestamp-grounded reasoning encourages the model to verify auditory evidence, moving toward mechanistic interpretability in LALMs.

6. Conclusion

In this work, we show that correct predictions in LALMs do not necessarily imply faithful grounding in acoustic evidence. To address this, we propose a timestamp-grounded reasoning framework that anchors intermediate reasoning steps to temporally localized audio segments, allowing both the model and users to access the acoustic evidence supporting each reasoning step. Experiments demonstrate that timestamp grounding improves benchmark performance, reasoning consistency, audio verification behavior, and interpretability. While our experiments focus on speech-centric tasks, out methodology is not conceptually limited to speech, and extending the framework to non-speech audio events is a plausible direction for future work.

7. Generative AI Use Disclosure

Generative AI tools were used only for minor language editing of the manuscript. All authors take responsibility for the content of this paper.

8. References

[1] Y. Chu, J. Xu, X. Zhou, Q. Yang, S. Zhang, Z. Yan, C. Zhou, and J. Zhou, "Qwen-audio: Advancing universal audio understanding via unified large-scale audio-language models," *arXiv preprint arXiv:2311.07919*, 2023.

[2] Z. Kong, A. Goel, R. Badlani, W. Ping, R. Valle, and B. Catanzaro, "Audio flamingo: a novel audio language model with few-shot learning and dialogue abilities," in *Proceedings of the 41st International Conference on Machine Learning*, 2024, pp. 25 125–25 148.

[3] C. Tang, W. Yu, G. Sun, X. Chen, T. Tan, W. Li, L. Lu, Z. MA, and C. Zhang, "Salmonn: Towards generic hearing abilities for large language models," in *The Twelfth International Conference on Learning Representations*, 2024.

[4] Y. Gong, A. H. Liu, H. Luo, L. Karlinsky, and J. Glass, "Joint audio and speech understanding," in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2023, pp. 1–8.

[5] S. Ghosh, S. Kumar, A. Seth, C. K. R. Evuru, U. Tyagi, S. Sakshi, O. Nieto, R. Duraiswami, and D. Manocha, "Gama: A large audio-language model with advanced audio understanding and complex reasoning abilities," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 6288–6313.

[6] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in neural information processing systems*, vol. 35, pp. 24 824–24 837, 2022.

[7] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in *The Eleventh International Conference on Learning Representations*, 2023.

[8] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in neural information processing systems*, vol. 35, pp. 22 199–22 213, 2022.

[9] S. C.-H. Yang, D. M. Wolpert, and M. Lengyel, "Theoretical perspectives on active sensing," *Current opinion in behavioral sciences*, vol. 11, pp. 100–108, 2016.

[10] K. J. Friston, N. Sajid, D. R. Quiroga-Martinez, T. Parr, C. J. Price, and E. Holmes, "Active listening," *Hearing research*, vol. 399, p. 107998, 2021.

[11] A. Jaech, A. Kalai, A. Lerer, A. Richardson, A. El-Kishky, A. Low, A. Helyar, A. Madry, A. Beutel, A. Carney *et al.*, "Openai o1 system card," *arXiv preprint arXiv:2412.16720*, 2024.

[12] D. Guo, D. Yang, H. Zhang, J. Song, P. Wang, Q. Zhu, R. Xu, R. Zhang, S. Ma, X. Bi *et al.*, "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025.

[13] K. Team, A. Du, B. Gao, B. Xing, C. Jiang, C. Chen, C. Li, C. Xiao, C. Du, C. Liao *et al.*, "Kimi k1. 5: Scaling reinforcement learning with llms," *arXiv preprint arXiv:2501.12599*, 2025.

[14] Y. Wang, P. Mousavi, A. Ploujnikov, and M. Ravanelli, "What are they doing? joint audio-speech co-reasoning," in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.

[15] J. Xu, Z. Guo, H. Hu, Y. Chu, X. Wang, J. He, Y. Wang, X. Shi, T. He, X. Zhu *et al.*, "Qwen3-omni technical report," *arXiv preprint arXiv:2509.17765*, 2025.

[16] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv *et al.*, "Qwen3 technical report," *arXiv preprint arXiv:2505.09388*, 2025.

[17] S. Ghosh, A. Goel, J. Kim, S. Kumar, Z. Kong, S.-g. Lee, C.-H. H. Yang, R. Duraiswami, D. Manocha, R. Valle *et al.*, "Audio flamingo 3: Advancing audio intelligence with fully open large audio language models," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.

[18] S. Deshmukh, S. Dixit, R. Singh, and B. Raj, "Mellow: a small audio language model for reasoning," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.

[19] Z. Ma, Z. Chen, Y. Wang, E. S. Chng, and X. Chen, "Audio-cot: Exploring chain-of-thought reasoning in large audio language model," *arXiv preprint arXiv:2501.07246*, 2025.

[20] Z. Xie, M. Lin, Z. Liu, P. Wu, S. Yan, and C. Miao, "Audio-reasoner: Improving reasoning capability in large audio language models," *arXiv preprint arXiv:2503.02318*, 2025.

[21] G. Li, J. Liu, H. Dinkel, Y. Niu, J. Zhang, and J. Luan, "Reinforcement learning outperforms supervised fine-tuning: A case study on audio question answering," *arXiv preprint arXiv:2503.11197*, 2025.

[22] A. Rouditchenko, S. Bhati, E. Araujo, S. Thomas, H. Kuehne, R. Feris, and J. Glass, "Omni-r1: Do you really need audio to fine-tune your audio llm?" *arXiv preprint arXiv:2505.09439*, 2025.

[23] S. Wu, C. Li, W. Wang, H. Zhang, H. Wang, M. Yu, and D. Yu, "Audio-thinker: Guiding audio language model when and how to think via reinforcement learning," *arXiv preprint arXiv:2508.08039*, 2025.

[24] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. Li, Y. Wu *et al.*, "Deepseekmath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024.

[25] Z. Liu, Z. Sun, Y. Zang, X. Dong, Y. Cao, H. Duan, D. Lin, and J. Wang, "Visual-rft: Visual reinforcement fine-tuning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 2034–2044.

[26] Q. Wu, X. Yang, Y. Zhou, C. Fang, B. Song, X. Sun, and R. Ji, "Grounded chain-of-thought for multimodal large language models," *arXiv preprint arXiv:2503.12799*, 2025.

[27] Y. Man, D.-A. Huang, G. Liu, S. Sheng, S. Liu, L.-Y. Gui, J. Kautz, Y.-X. Wang, and Z. Yu, "Argus: Vision-centric reasoning with grounded chain-of-thought," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 14 268–14 280.

[28] G. H. Sarch, S. Saha, N. Khandelwal, A. Jain, M. J. Tarr, A. Kumar, and K. Fragkiadaki, "Grounded reinforcement learning for visual reasoning," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.

[29] F. Tian, X. T. Zhang, Y. Zhang, H. Zhang, Y. Li, D. Liu, Y. Deng, D. Wu, J. Chen, L. Zhao *et al.*, "Step-audio-r1 technical report," *arXiv preprint arXiv:2511.15848*, 2025.

[30] S. Kumar, P. Seetharaman, K. Chen, O. Nieto, J. Su, Z. Wang, R. Kumar, D. Manocha, N. J. Bryan, Z. Jin *et al.*, "Tac: Timestamped audio captioning," *arXiv preprint arXiv:2602.15766*, 2026.

[31] J. Louradour, "whisper-timestamped," https://github.com/linto-ai/whisper-timestamped, 2023.

[32] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *International conference on machine learning*, 2023, pp. 28 492–28 518.

[33] T. Giorgino, "Computing and visualizing dynamic time warping alignments in r: The dtw package," *Journal of Statistical Software*, vol. 31, no. 7, 2009.

[34] R.-G. Dumitru, D. Peteleaza, V. Yadav, and L. Pan, "Conciserl: Conciseness-guided reinforcement learning for efficient reasoning models," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 17 099–17 123.

[35] L. Yue, Y. Du, Y. Wang, W. Gao, F. Yao, L. Wang, Y. Liu, Z. Xu, Q. Liu, S. Di *et al.*, "Don't overthink it: A survey of efficient r1-style large reasoning models," *arXiv preprint arXiv:2508.02120*, 2025.

[36] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: an asr corpus based on public domain audio books," in *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2015, pp. 5206–5210.

[37] C. Wang, A. Wu, and J. Pino, "Covost 2 and massively multilingual speech-to-text translation," *arXiv preprint arXiv:2007.10310*, 2020.

[38] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "Meld: A multimodal multi-party dataset for emotion recognition in conversations," in *Proceedings of the 57th annual meeting of the association for computational linguistics*, 2019, pp. 527–536.

[39] S. Abu-El-Haija, N. Kothari, J. Lee, P. Natsev, G. Toderici, B. Varadarajan, and S. Vijayanarasimhan, "Youtube-8m: A large-scale video classification benchmark," *arXiv preprint arXiv:1609.08675*, 2016.

[40] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang, B. Zhang, X. Wang, Y. Chu, and J. Lin, "Qwen2.5-omni technical report," *arXiv preprint arXiv:2503.20215*, 2025.

[41] P. Pezeshkpour, M. Aminnaseri, and E. Hruschka, "Mixed signals: Decoding vlms' reasoning and underlying bias in vision-language conflict," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 24 833–24 848.

[42] A. Vo, K.-N. Nguyen, M. R. Taesiri, V. T. Dang, A. T. Nguyen, and D. Kim, "Vision language models are biased," in *Proceedings of the Fourteenth International Conference on Learning Representations*, 2026.

[43] S. Sakshi, U. Tyagi, S. Kumar, A. Seth, R. Selvakumar, O. Nieto, R. Duraiswami, S. Ghosh, and D. Manocha, "Mmau: A massive multi-task audio understanding and reasoning benchmark," in *International Conference on Learning Representations*, 2025, pp. 84 929–84 964.

[44] G. Comanici, E. Bieber, M. Schaekermann, I. Pasupat, N. Sachdeva, I. Dhillon, M. Blistein, O. Ram, D. Zhang, E. Rosen *et al.*, "Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities," *arXiv preprint arXiv:2507.06261*, 2025.

[45] A. Mesaros, T. Heittola, and T. Virtanen, "Metrics for polyphonic sound event detection," *Applied Sciences*, vol. 6, no. 6, p. 162, 2016.

[46] J. Lindsey, W. Gurnee, E. Ameisen, B. Chen, A. Pearce, N. L. Turner, C. Citro, D. Abrahams, S. Carter, B. Hosmer, J. Marcus, M. Sklar, A. Templeton, T. Bricken, C. McDougall, H. Cunningham, T. Henighan, A. Jermyn, A. Jones, A. Persic, Z. Qi, T. B. Thompson, S. Zimmerman, K. Rivoire, T. Conerly, C. Olah, and J. Batson, "On the biology of a large language model," *Transformer Circuits Thread*, 2025.

[47] O. Skean, M. R. Arefin, D. Zhao, N. N. Patel, J. Naghiyev, Y. Lecun, and R. Shwartz-Ziv, "Layer by layer: Uncovering hidden representations in language models," in *International Conference on Machine Learning*, 2025, pp. 55 854–55 875.

[48] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford *et al.*, "Gpt-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.

[49] Z. Ma, Y. Ma, Y. Zhu, C. Yang, Y.-W. Chao, R. Xu, W. Chen, Y. Chen, Z. Chen, J. Cong *et al.*, "Mmar: A challenging benchmark for deep reasoning in speech, audio, music, and their mix," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2025.

[50] Q. Yang, J. Xu, W. Liu, Y. Chu, Z. Jiang, X. Zhou, Y. Leng, Y. Lv, Z. Zhao, C. Zhou *et al.*, "Air-bench: Benchmarking large audio-language models via generative comprehension," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024, pp. 1979–1998.